Effectiveness of Sitagliptin and Metformin FDC in Indian Patients with T2DM from a Real-world Retrospective EMR Based Study

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Introduction

The "Real Burden"

- Every 6th person with diabetes in the world is an Indian.[1]
- The 2021 ICMR-INDIAB study reported that the prevalence of diabetes in India stands at 101 million, highlighting the extensive burden of these conditions on the population.[2]
- In diabetic patients there is a high prevalence of overweight and obese with some studies reporting the prevalence to be as high as 80 -90 %.[3,4]

Guidelines

- The American Diabetes Association 2024 guidelines recommend: For achieving glycaemic goals choose metformin or other agents or combination therapy to provide adequate efficacy and maintain treatment goals.[5]

Synergistic Effect of Sitagliptin + Metformin FDC

- Due to the progressive decline in β-cell function, it is often necessary to utilise multiple agents with complementary mechanism of action to address multiple core pathophysiologies in T2DM to achieve optimal glycaemic control.[6-8]
- When used together, sitagliptin and metformin complement each other's mechanism of action, leading to better glycaemic control than either medication alone.[7]
- Metformin reduces blood glucose levels by decreasing glucose production in the liver, and increasing insulin sensitivity while sitagliptin inhibits dipeptidyl peptidase-4 (DPP-4), in turn increases glucagon-like peptide-1 (GLP-1), results in increased insulin release and improved glucose tolerance.[7]
- Sitagliptin + metformin combination have shown a doubling of active postprandial GLP-1 levels, which is 4-fold greater than metformin or sitagliptin alone.[6]
- Sitagliptin and its FDCs have low risk of hypoglycaemia, provide β-cell protection, and is cardiorenal safe.[6-8]

Synergistic Effect of Sitagliptin + Metformin FDC that Addresses Multiple Pathophysiologies in T2DM[4-6]

- **DPP4i** increases GLP-1 and GIP levels by 2-fold, thus stimulating insulin secretion, inhibits glucagon secretion, enhances insulin release, brings about β-cell preservation and is weight neutral.
- **Metformin** lowers hepatic glucose output, improves β-cell response and function, improves insulin resistance, increases liver and peripheral tissue sensitivity to glucose and is weight neutral.

Objective

- A real-world, retrospective, observational, electronic medical records (EMR) based study was conducted to understand the effectiveness of sitagliptin and sitagliptin + metformin fixed dose combination (FDC) in Indian patients with type 2 diabetes mellitus (T2DM). The study was approved by Independent Ethics Committee and registered on CTRI (CTRI/2023/10/058366).

Materials and Methods

- The EMR data of adults aged ≥ 18 years of either gender having T2DM, between 2017 and 2023 who were prescribed sitagliptin or sitagliptin + metformin FDC, with or without other OHAs, with at-least 1 follow-up visit at 3 months was retrieved. The data here presents a subgroup analysis of patients who received sitagliptin + metformin FDC and assessed the effectiveness of sitagliptin + metformin FDC on glycosylated haemoglobin (HbA1c), fasting blood glucose (FBG) and postprandial blood glucose (PPBG) levels in total pool of patients from baseline to 3 months with HbA1c ≥ 7% at baseline.
- **Primary end-points**
  - Mean change in HbA1c from baseline to 3 months in patients with HbA1c ≥ 7% at baseline
- **Secondary end-points**
  - Mean change in PPBG and FBG from baseline to 3 months in patients with baseline HbA1c ≥ 7%
  - Proportion of patients achieving HbA1c< 7%

Results

- In the current EMR data, 5,721 patients were prescribed with sitagliptin + metformin of which 2,113 had HbA1c ≥ 7% at baseline.

Glycosylated haemoglobin

A statistically significant reduction in HbA1c was observed from baseline to 3 months [(8.57 ± 1.50 to 7.54 ± 1.11; Difference of -1.03 ± 1.59, $p < 0.001$, N=2,113) Figure 1].

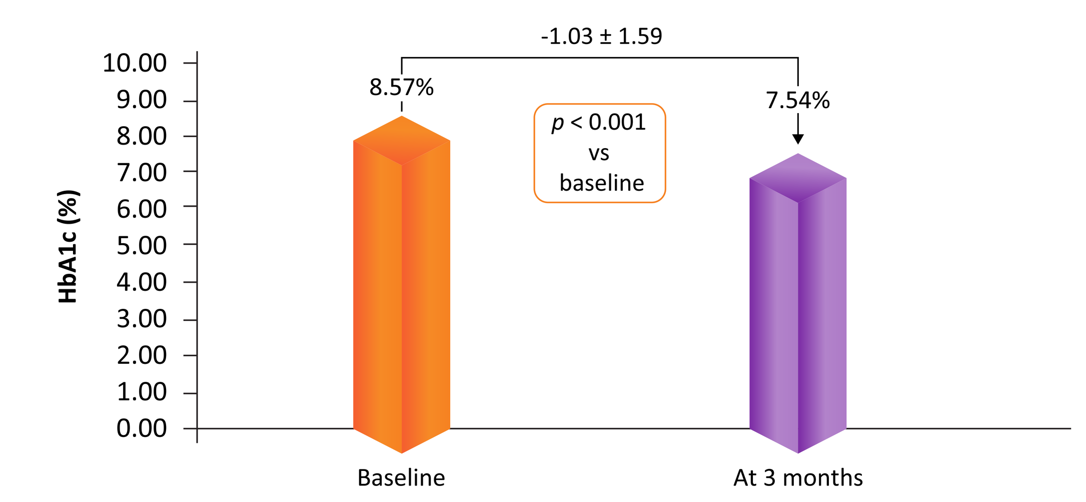

Figure 1: Mean change in HbA1c from baseline to 3 months (N = 2,113)

Fasting blood glucose levels

A statistically significant reduction in FBG was observed from baseline to 3 months [(161.57 ± 53.14 to 135.12 ± 40.04, $p < 0.001$, N = 1,928) Figure 2].

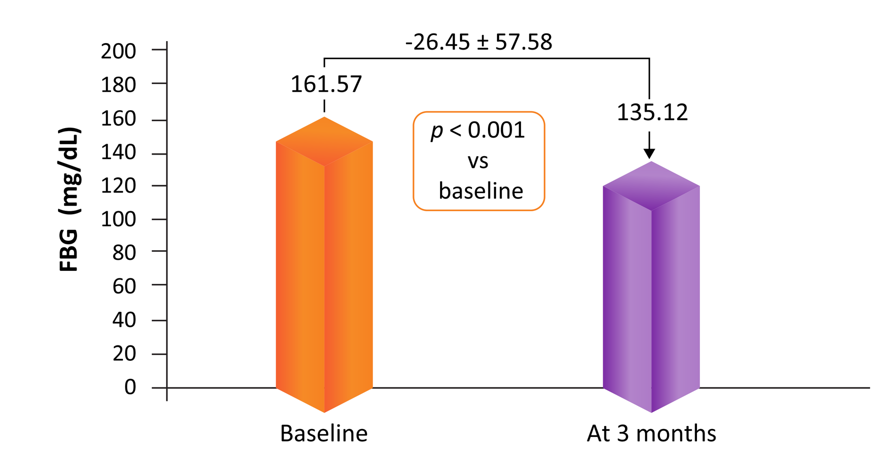

Figure 2: Mean change in FBG from baseline to 3 months (N = 1,928)

Postprandial blood glucose levels

A statistically significant reduction in PPBG was observed from baseline to 3 months [(232.62 ± 76.97 to 192.55 ± 63.22, $p < 0.001$, N = 1,605) Figure 3].

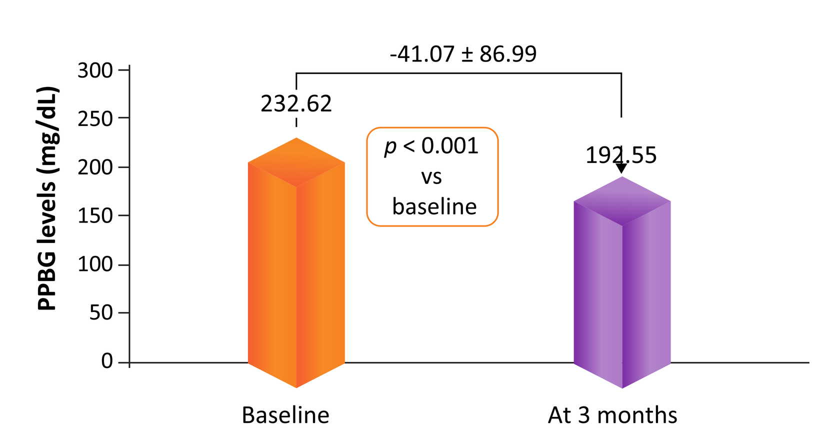

Figure 3: Mean change in PPBG from baseline to 3 months (N = 1,605)

- There was statistically significant improvement with sitagliptin + metformin FDC in the HbA1c, FBG and PPBG at the end of 3 months ($p < 0.001$).
- At the end of 3 months 665 patients (31.47%) achieved HbA1c < 7%.

Limitation

This was a single arm analysis of EMR data captured on digital platform and the database that we used, does not capture safety data

Conclusion

- This real-world, EMR-based study in India demonstrated effectiveness of sitagliptin and metformin FDC, in ters of significant improvement in HbA1c, FBG, and PPBG in treatment of patients with T2DM having baseline HbA1c ≥7%.
- Sitagliptin and metformin combination offers a complementary mechanism of action that enhances glycaemic control, particularly beneficial for Indian patients by addressing insulin resistance and reducing hepatic glucose production.
- A substantial proportion of patients achieved HbA1c levels < 7% after 3 months of treatment with sitagliptin and metformin FDC, highlighting its effectiveness in real-world settings as a therapeutic option for managing T2DM and achieving glycaemic targets.
- Long term studies are needed to assess effectiveness and safety of sitagliptin and metformin combination.

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Abbreviations: EMR, electronic medical records; FBG, fasting blood glucose; FDC, fixed-dose combination; GLP-1, glucagon-like peptide-1; HbA1c, glycosylated haemoglobin; ICMR-INDIAB, Indian Council of Medical Research-India Diabetes; DPP4i, dipeptidyl peptidase 4 inhibitor; PPBG, postprandial blood glucose; T2DM, type 2 diabetes mellitus.

References
1. Mohan V, Mukherjee JJ, Das AK, *et al. Endocrine and Metabolic Science.* 2021 Sep 30;4:100103. **2.** Anjana RM, Unnikrishnan R, Deepa M, *et al.* Metabolic non-communicable disease health report of India: the ICMR-INDIAB national cross-sectional study (ICMR-INDIAB-17). *The Lancet Diabetes Endocrinol. 2023;11(7):474-489.* **3.** American Diabetes Association Professional Practice Committee. 9. Pharmacologic Approaches to Glycemic Treatment: Standards of Care in Diabetes—2024. *Diabetes Care* 2024;47(Suppl. 1):S158–S178 **4.** Hayes J, Anderson R, Stephens JW. Sitagliptin/metformin fixed-dose combination in type 2 diabetes mellitus: an evidence-based review of its place in therapy. *Drug Des Devel Ther. 2016;10:2263-2270.* **5.** Aashish M, Arindam N, Siddiqi SS, *et.al.* Efficacy and safety of fixed dose combination of Sitagliptin, metformin, and pioglitazone in type 2 Diabetes (IMPACT study): a randomized controlled trial. *Clin Diabetes Endocrinol. 2024;10(1):3.* **6.** Goldstein BJ, Feinglos MN, Lunceford JK, *et al.* Effect of initial combination therapy with sitagliptin, a dipeptidyl peptidase-4 inhibitor, and metformin on glycemic control in patients with type 2 diabetes. *Diabetes Care. 2007;30(8):1979-1987.*

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